

Assignment 6: Sobol and Dynamic Model

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The following document contains responses to Assignment 6 of EDS230: Modeling Environmental Systems. It considers a model of forest growth, where forest size is measured in units of carbon (C).

The three variable parameters include K (carrying capacity), r (exponential growth rate before canopy closure), and g (linear growth rate after canopy closure). There is a canopy closure threshold of 50 kg/C, where the rate of forest growth changes. The forest model function can be found within the R folder of this repository, called `dforestgrowth`.

0. Load necessary libraries

```
# libraries
library(tidyverse)
library(deSolve)
library(here)
library(sensitivity)
```

1. Implement Model in R as Differential Equations

```
# forest growth function
source(here("R/dforestgrowth.R"))
```

2. Run the Model for 300 years using ODE solver and defined parameters

```

# create a sequence of years from 1 to 300
time <- seq(from = 1, to = 300)

# initial forest size
Cinitial <- 10

# create list of parameters to input into ode
params <- list(K = 250, r = 0.01, g = 2)

# run ode function and save result
res <- ode(y = Cinitial, times = time, func = dforestgrowth, parms = params)

# rename columns for clarity
colnames(res) <- c("time", "C")

```

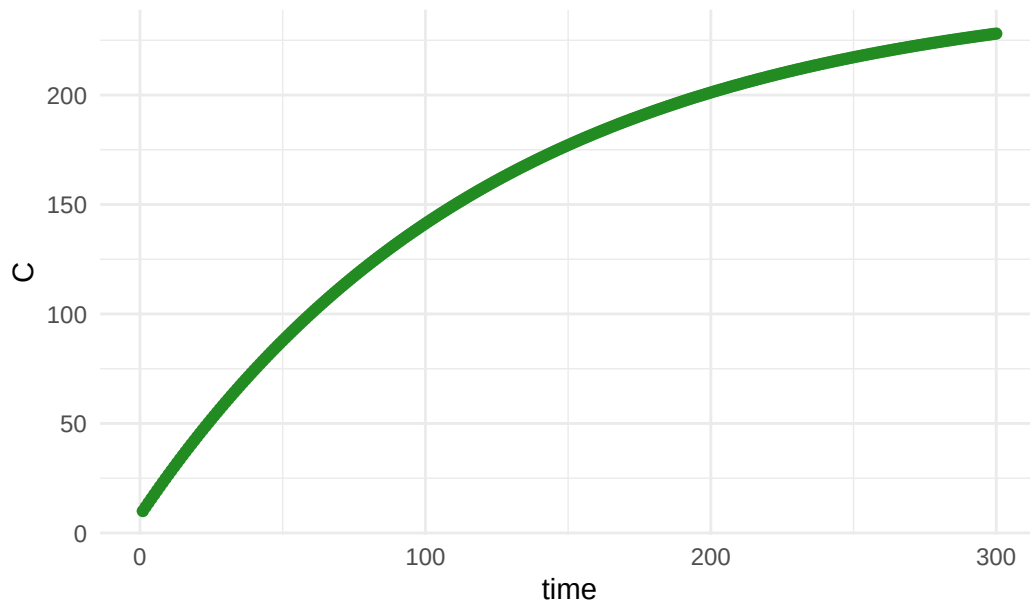
3. Graph ODE results

```

# plot
ggplot(as.data.frame(res), aes(time, C)) +
  geom_point(color= "forestgreen") +
  labs(title = "Forest size trajectory with no uncertainty") +
  theme_minimal()

```

Forest size trajectory with no uncertainty



4. Run a Sobol Global

```
# create wrapper function
c_wrapper <- function(K, r, g, Cinitial, simtimes, odefunc, metricfunc) {

  # create parameter list
  params <- list(K = K, r = r, g = g)

  # run ode
  result <- ode(y = Cinitial, times = simtimes, func = odefunc, parms = params, method="euler")

  # results as a data frame
  result = data.frame(time=result[,1], dC=result[,2])

  # compute metrics
  metrics <- metricfunc(as.data.frame(result))
  return(metrics)
}

# create function to compute metric, which is maximum forest size
compute_growth_metric <- function(result) {
```

```

# first add maximum population
maxforestsize <- max(result$dC)

# return results
return(list(maxforestsize = maxforestsize))
}

```

```

# test wrapper function using specific values
c_wrapper(K = 250, r = 0.01, g = 2, Cinitial = 10,
          simtimes = seq(from = 1, to = 300),
          odefunc = dforestgrowth, metricfunc = compute_growth_metric)

```

```

$maxforestsize
[1] 228.2629

```

```

# create parameter distributions to test sensitivity across 3 variables

# set number of parameters
np <- 2000

# create first set of normal distributions of K, r, and g
K <- rnorm(mean = 250, sd = 0.1*250, n = np)
r <- rnorm(mean = 0.01, sd = 0.1 * 0.01, n = np)
g <- rnorm(mean = 2, sd = 0.1 * 2, n = np)
x1 <- cbind.data.frame(K = K, r = r, g = g)

# create second set of normal distributions
K <- rnorm(mean = 250, sd = 0.1*250, n = np)
r <- rnorm(mean = 0.01, sd = 0.1 * 0.01, n = np)
g <- rnorm(mean = 2, sd = 0.1 * 2, n = np)
x2 <- cbind.data.frame(K = K, r = r, g = g)

# fix any negative values and they are not meaningful
# the pmax function makes any values that are negative 0
x1 <- x1 %>% map_df(pmax, 0.0)
x2 <- x2 %>% map_df(pmax, 0.0)

# create Sobol object and get the parameter set we will be running the model on
sens_C <- sobolSalt(model = NULL, x1, x2, nboot = 300)

# assign proper column names for interpretability

```

```
colnames(sens_C$X) <- c("K", "r", "g")
```

```
# view parameter set  
head(sens_C$X)
```

```
      K      r      g  
[1,] 221.7282 0.010590919 2.139695  
[2,] 300.9805 0.008181983 2.220982  
[3,] 257.9717 0.010850356 2.055274  
[4,] 191.7076 0.008465395 2.048753  
[5,] 276.1216 0.009501285 2.169336  
[6,] 214.2337 0.010283746 2.018308
```

```
# run wrapper on all parameter sets  
allresults <- as.data.frame(sens_C$X) %>%  
  pmap(c_wrapper, Cinitial = 10, simtimes = seq(from = 1, to = 300),  
       odefunc = dforestgrowth, metricfunc = compute_growth_metric)
```

```
# extract out results from pmap into a data frame  
allres <- allresults %>%  
  map_dfr(`[, c("maxforestsize")])
```

```
# do a quick summary to get a sense of the range  
summary(allres)
```

```
maxforestsize  
Min.      :154.1  
1st Qu.:214.3  
Median :226.9  
Mean    :226.5  
3rd Qu.:238.7  
Max.    :286.8
```

5. Graph sensitivity of max forest size as a box plot and record Sobol indices

```
# extract Sobol indices S and T  
sens_C <- tell(sens_C, allres$maxforestsize)  
  
# rename row names for clarity
```

```
rownames(sens_C$S) <- c("K", "r", "g")
rownames(sens_C$T) <- c("K", "r", "g")

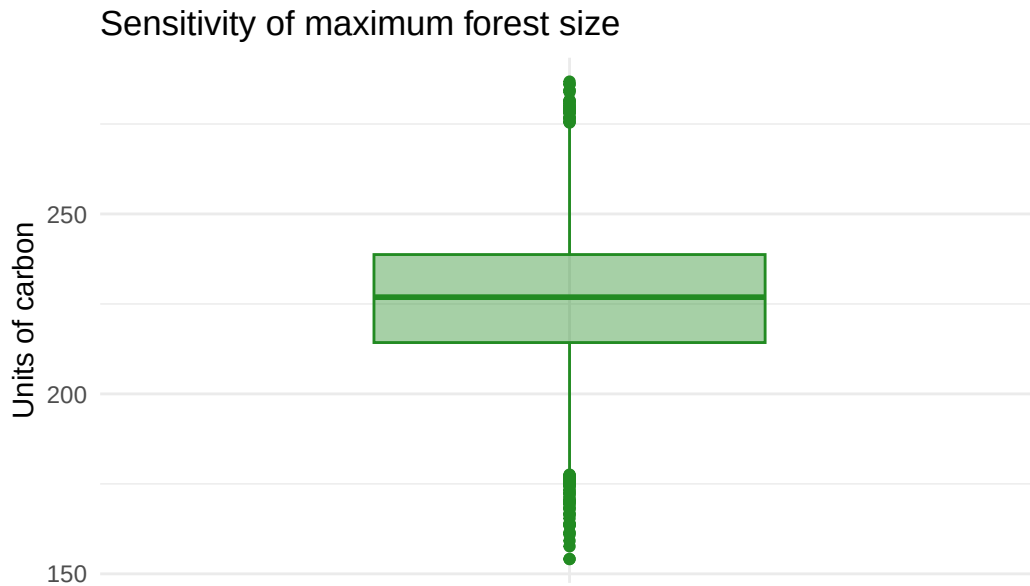
print(sens_C$S)
```

	original	bias	std. error	min. c.i.	max. c.i.
K	0.90670228	0.0003849928	0.004391134	0.89809736	0.91469968
r	0.02531939	-0.0011913174	0.022531771	-0.01568093	0.07697525
g	0.09103138	-0.0013884925	0.022154683	0.05152113	0.13724837

```
print(sens_C$T)
```

	original	bias	std. error	min. c.i.	max. c.i.
K	8.763907e-01	1.029017e-03	2.111601e-02	8.290039e-01	9.164544e-01
r	-1.297629e-12	8.773945e-13	9.013562e-13	-3.930826e-12	-4.881405e-13
g	9.467369e-02	-1.263063e-04	4.954647e-03	8.473511e-02	1.039940e-01

```
# boxplot of maximum forest size
tmp <- allres %>% pivot_longer(cols = everything(), names_to = "metric", values_to = "value")
ggplot(allres, aes(x = " ",
                   y = maxforestsize)) +
  geom_boxplot(width = 0.5,
               color = "forestgreen",
               fill = alpha("forestgreen", 0.4)) +
  labs(x = " ",
       title = "Sensitivity of maximum forest size",
       y = "Units of carbon") +
  theme_minimal()
```



6. In 2-3 sentences, discuss what the results of your simulation might mean

From the first-order Sobol indices, we can see that with no consideration of co-variance between parameters, K , or carrying capacity, has the largest effect on maximum forest size. From the total sensitivity index (or T), we see a similar pattern, with carrying capacity contributing the most – by far – to maximum forest size over 300 years.

These results suggest that climate change's influence on carrying capacity could significantly impact maximum forest size. Variables that impact carrying capacity include climate, water availability, nutrients, and space/habitat, all of which climate change affects in the natural environment.